

Selective recruitment: Evidence for task-dependent gating of inputs to the cerebellum

Reviewed Preprint

v2 • June 13, 2024

Revised by authors

Reviewed Preprint

v1 • April 5, 2024

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Abstract

Summary

fMRI studies have documented cerebellar activity across a wide array of tasks. However, the functional contribution of the cerebellum within these task domains remains unclear because cerebellar activity is often studied in isolation. This is problematic, as cerebellar fMRI activity may simply reflect the transmission of neocortical activity through fixed connections. Here we present a new approach that addresses this problem. Rather than focus on task-dependent activity changes in the cerebellum alone, we ask if neocortical inputs to the cerebellum are gated in a task-dependent manner. We hypothesize that input is upregulated when the cerebellum functionally contributes to a task. We first validated this approach using a finger movement task, where the integrity of the cerebellum has been shown to be essential for the coordination of rapid alternating movements but not for force generation. While both neocortical and cerebellar activity increased with increasing speed and force, the speed-related changes in the cerebellum were larger than predicted by an optimized cortico-cerebellar connectivity model. We then applied the same approach in a cognitive domain, assessing how the cerebellum supports working memory. Enhanced gating was associated with the encoding of items in working memory, but not with the manipulation or retrieval of the items. Focusing on task-dependent gating of neocortical inputs to the cerebellum offers a promising approach for using fMRI to understand the specific contributions of the cerebellum to cognitive function.

eLife assessment

This **important** study reports a novel approach to studying cerebellar function based on the idea of selective recruitment using fMRI. It provides **convincing** evidence for task-dependent gating of neocortical input to the cerebellum during a motor task and a working memory task. The study will be of interest to a broad cognitive neuroscience audience.

<https://doi.org/10.7554/eLife.96386.2.sa2>

Introduction

More than 30 years of neuroimaging has revealed that the human cerebellum is activated in a broad range of tasks including motor (Spraker et al., 2012 [↗](#)), language (Petersen et al., 1989 [↗](#)), working memory (Marvel and Desmond, 2010 [↗](#)), attention (Allen et al., 1997 [↗](#)), social (Van Overwalle et al., 2015 [↗](#)), and visual cognition tasks (van Es et al., 2019 [↗](#)) - for a review see (Diedrichsen et al., 2019 [↗](#)). Indeed, there are very few tasks that do not lead to activity in some part of the cerebellum. The presence of cerebellar activity is usually taken as evidence that the cerebellum plays a functional role associated with these tasks.

However, there is an important problem with this line of reasoning. The cerebellar BOLD signal does not reflect the activity levels of Purkinje cells, the output of the cerebellar cortex (Caesar et al., 2003 [↗](#); Thomsen et al., 2004 [↗](#), 2009 [↗](#)). Rather, it is determined solely by mossy fiber (Akgören et al., 1994 [↗](#); Gagliano et al., 2022 [↗](#); Mapelli et al., 2017 [↗](#)) and climbing fiber (Caesar et al., 2003 [↗](#); Mathiesen et al., 2000 [↗](#)) input, with the former likely playing the dominant role (Attwell and Iadecola, 2002 [↗](#); Howarth et al., 2012 [↗](#)).

Mossy fibers carry input from a wide array of neocortical areas, including prefrontal and parietal association cortices, as demonstrated directly through viral tracing studies in non-human primates (Kelly and Strick, 2003 [↗](#)), and indirectly through resting-state functional connectivity (rs-FC) analysis in humans (Buckner et al., 2011 [↗](#); Ji et al., 2019 [↗](#); Marek et al., 2018 [↗](#); O'Reilly et al., 2010 [↗](#)). This means that increases in the cerebellar BOLD signal could simply reflect the automatic transmission of neocortical activity through fixed anatomical connections. As such, whenever a task activates a neocortical region, the corresponding cerebellar region would also be activated, regardless of whether the cerebellum is directly involved in the task or not.

The preceding arguments suggest that it is important to consider cerebellar activation in the context of the neocortical regions that provide its input. To approach this problem, we have recently developed and tested a range of cortical-cerebellar connectivity models (King et al., 2023 [↗](#)), designed to capture fixed, or task-invariant, transmission between neocortex and cerebellum. For each cerebellar voxel, we estimated a regularized multiple regression model to predict its activity level across a range of task conditions (King et al., 2019 [↗](#)) from the activity pattern observed in the neocortex for the same conditions. The models were then evaluated in their ability to predict cerebellar activity in novel tasks, again based only on the corresponding neocortical activity pattern. Two key results emerged from this work. First, while rs-FC studies (Buckner et al., 2011 [↗](#); Ji et al., 2019 [↗](#); Marek et al., 2018 [↗](#)) have assumed a 1:1 mapping between neocortical and cerebellar networks, models which allowed for convergent input from multiple neocortical regions to a single cerebellar region performed better in predicting cerebellar

activity patterns. Second, when given a cortical activation pattern, the best performing model could predict about 50% of the reliable variance in the cerebellar cortex across tasks (King et al., 2023 [↗](#)).

This model offers a powerful null model to evaluate whether the cerebellar BOLD signal can be fully explained by the fixed transmission of input from neocortex in a task-invariant manner. The fact that the prediction of these models did not reach the theoretically possible prediction accuracy suggests that the connectivity between the neocortex and cerebellum may not be fully task-invariant. Instead, neocortical input to the cerebellum may be modulated as a function of the relative importance of cerebellar computation in a task-specific manner. We refer to this as the **selective recruitment** hypothesis; specifically, we hypothesize that input is upregulated when cerebellar computation is required. Such task- or state-dependent gating would make evolutionary sense, given the substantial metabolic cost of granule cell activity (Attwell and Iadecola, 2002 [↗](#); Howarth et al., 2010 [↗](#)).

To evaluate the selective recruitment hypothesis, we first turned to the motor domain where clinical studies provide a strong *a priori* hypothesis of when the cerebellum should be selectively recruited. Patients with cerebellar damage consistently show impairments in performing rapid alternating movements, a symptom called dysdiadochokinesia (Hallett et al., 1991 [↗](#); Mai et al., 1988 [↗](#)). In contrast, these patients are generally able to exert grip forces comparable to healthy controls (Mai et al., 1988 [↗](#)). Based on the selective recruitment hypothesis, we predicted that increases in cerebellar BOLD will be greater for increases in tapping speed compared to increases in finger force output, even when the neocortical activity is matched between conditions.

We then applied the approach in a cognitive domain. Working memory tasks have been shown to robustly activate hemispheric regions of cerebellar lobules VI, VII, and VIII (Chen and Desmond, 2005 [↗](#); Desmond et al., 1997 [↗](#)). Furthermore, patients with cerebellar damage tend to show deficits in verbal working memory tasks (Cooper et al., 2012 [↗](#); Ilg et al., 2013 [↗](#); Kansal et al., 2017 [↗](#); Peterburs et al., 2010 [↗](#); Ravizza et al., 2006 [↗](#)). However, the form of the deficits is unclear and quite variable (Hokkanen et al., 2006 [↗](#); McDougle et al., 2022 [↗](#); Pleger and Timmann, 2018 [↗](#); Starowicz-Filip et al., 2021 [↗](#)), making it difficult to draw inferences concerning the computational contribution of the cerebellum to working memory tasks. We designed a digit-span task that allowed us to evaluate three factors relevant for working memory: 1) task phase (encoding/recall), 2) memory load, and 3) information manipulation (forward/backward recall). We asked which combination of these three factors leads to selective recruitment in cerebellar working memory regions.

Results

Motor task

To test the selective recruitment hypothesis in the motor domain, we used a task which involved alternating finger presses of middle and ring finger (**Figure 1A** [↗](#)). Starting at a baseline level of 1Hz and 2.5N, we either increased the force of each response or the required rate (**Table 1** [↗](#)). Both manipulations are expected to produce an increase in the BOLD response in neocortical motor areas (Diedrichsen et al., 2013 [↗](#); Thickbroom et al., 1998 [↗](#)). As such, our task-invariant connectivity model predicts increased cerebellar activity with both increases in speed and force (Spraker et al., 2012 [↗](#)). Critically, selective recruitment predicts that for equivalent activity levels in the neocortex, cerebellar activity should be higher in the speed than in the force condition.

Participants complied well with task instructions, as evidenced by the group-averaged peak forces and number of taps, which were close to the target values (**Table 1** [↗](#)). The high error rate for the baseline condition reflects the fact that some of the participants completed the 6 taps in less than

Table 1.

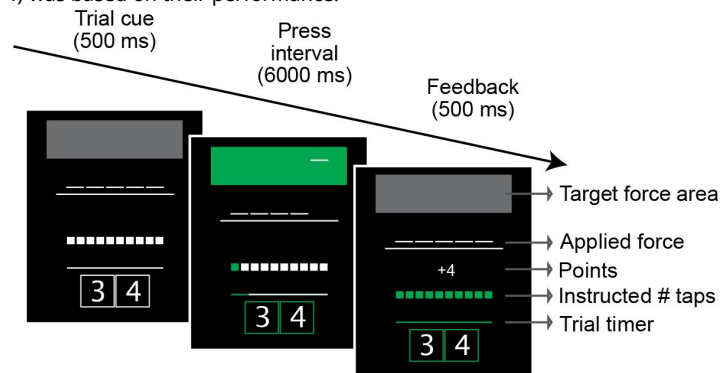
Mean and between-subject standard deviation ($\pm$) of force, speed, and error rate for each condition across subjects.

Condition	Target Force (N)	Target # taps in 6s	Average force (N)	Average # taps in 6s	Error rate (%)
High Speed	2.5	18	2.93 ± 0.48	17.72 ± 0.84	5 ± 0.21
Medium Speed	2.5	10	2.84 ± 0.45	10.12 ± 0.44	1 ± 0.12
Baseline	2.5	6	2.80 ± 0.41	6.32 ± 0.8	15 ± 0.36
Medium Force	6	6	6.10 ± 0.49	6.04 ± 0.2	4 ± 0.18
High Force	10	6	9.73 ± 0.66	6.04 ± 0.2	2 ± 0.13

Figure 1.

Timeline of events in the alternating finger tapping task.

The height of the target force area indicated the target force, the number of white squares the target number of taps. During the press interval, the participant alternatively tapped the middle and ring finger. After each tap, the next box turned green. Reward feedback (e.g., +4) was based on their performance.



the minimum interval of 4s in this very easy condition.

Increasing force and speed leads to increased activation in cortico-cerebellar motor network

As expected for right-hand movements, activation was observed in the hand areas of left (contralateral) M1 and S1 (**Figure 2**). Compared to the baseline condition, the combined M1/S1 ROI showed a significant activation increase in the high-force ($t_{15} = 9.41$, $p = 1.10 \times 10^{-7}$) and the high-speed conditions ($t_{15} = 8.29$, $p = 5.54 \times 10^{-7}$). Similarly, activity in the right anterior and posterior motor areas of the cerebellum (outlined in light gray in **Figure 2**, see *Methods* for details on region of interest) increased with increasing force ($t_{15} = 14.21$, $p = 4.14 \times 10^{-10}$) and speed ($t_{15} = 7.60$, $p = 1.59 \times 10^{-6}$). The medium force and speed conditions were between baseline and high conditions, replicating previous findings of a parametric modulation of activity with both force (Spraker et al., 2012) and speed (Jäncke et al., 1999).

Visual inspection of the activation maps (**Figure 2D** vs. **2F**) suggests that cerebellar activity increased more with speed than with force. One might take this result alone as an indication that recruitment of the cerebellum is relatively greater when the task requires the coordination of rapid finger movements compared to when an increase in force is required. However, the neocortical activation patterns for speed and force conditions were not completely matched (**Figure 2A** vs. **2C**): Increasing speed led to more widespread activation in secondary motor areas compared to increasing force. Therefore, the observed differences in cerebellar activity could have resulted from additional fixed inputs from premotor and supplementary motor areas, rather than from a task-dependent recruitment of cerebellar circuits for the speed task.

Cerebellar activity for increased speed is larger than predicted by task-invariant connectivity

To distinguish these two hypotheses, we used our task-invariant cortico-cerebellar connectivity model (L2-regularized multiple regression, see methods), trained on a separate set of participants across a large range of tasks (King et al., 2023). This model provides an estimate of cerebellar activity expected from fixed anatomical connections with the neocortex. We take this as the reference point for asking if observed activation levels are greater than expected; what we use as our operational definition of selective recruitment. **Figure 3A** shows the connectivity weights from this model for the cerebellar right hand area, region M3 (Nettekoven et al., 2023). According to the model, inputs to cerebellar M3 do not only come from contralateral M1 and S1, but also from premotor and supplementary motor regions.

We multiplied the neocortical activity patterns from each individual and condition with the connectivity weights from the model to predict the corresponding cerebellar M3 activity level. Note that the connectivity weights were estimated on subjects from independent task-based fMRI datasets; therefore, the predicted values were on a different scale compared to the observed values (**Figure 3B**). To account for this scaling difference, we used a simple linear regression between observed and predicted values.

In general, the predicted values closely match the observed values (average $R^2 = 0.60$, $SEM = 0.01$). However, relative to the force conditions, the speed conditions resulted in larger cerebellar activity, even though the predicted activity was smaller. To test for systematic deviations across subjects, we submitted the signed residuals for all conditions to a one-way ANOVA, revealing a significant effect of condition ($F_{4, 60} = 6.796$, $p = 1.1 \times 10^{-4}$). Post-hoc tests revealed that the signed residual for the high-speed condition was significantly higher than for the high force condition ($t_{15} = 2.37$, $p = 0.0157$). This was also the case when comparing medium speed and medium force ($t_{15} =$

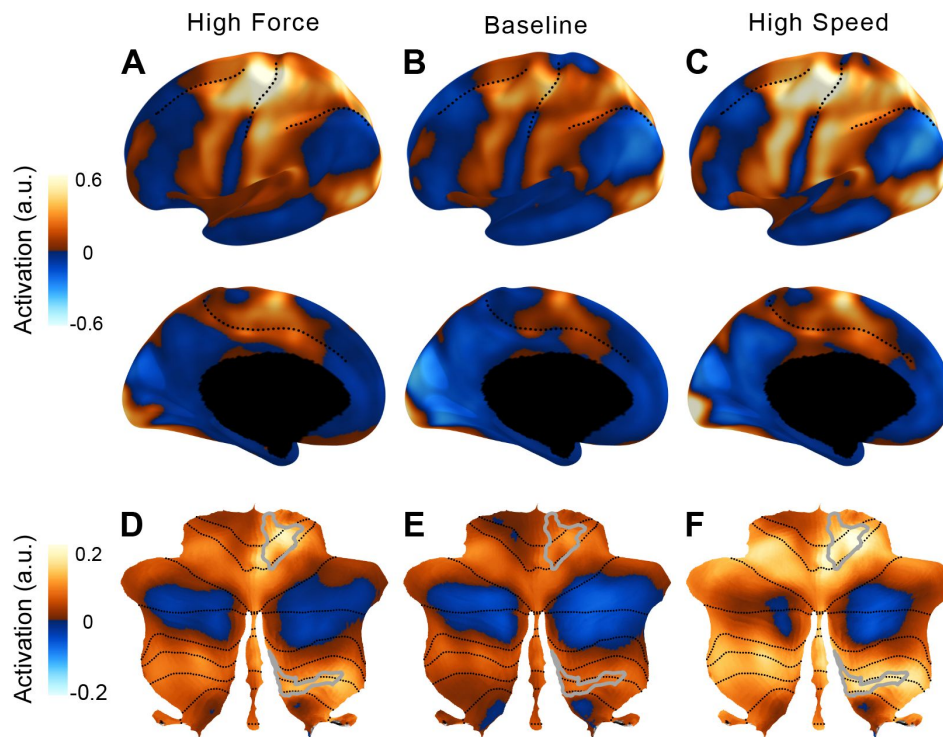


Figure 2.

Activation in the cortico-cerebellar motor network compared to rest.

Activity maps for high force (left), baseline (middle), and high speed (right) conditions. High levels of force and speed were chosen to show the spatial distribution of activity. Medium level of force and speed resulted in similar maps with activity levels between the baseline and high conditions. (A-C) Lateral and medial surface of the left hemisphere. Dotted lines indicate the superior frontal, central, intra-parietal and cingulate sulcus. (D-F) Flat map of the cerebellum (Diedrichsen and Zotow, 2015) with lobular boundaries indicated in dotted line. The right anterior and posterior hand motor area (gray outline) was defined by a new functional atlas of the cerebellum (Nettekoven et al., 2023).

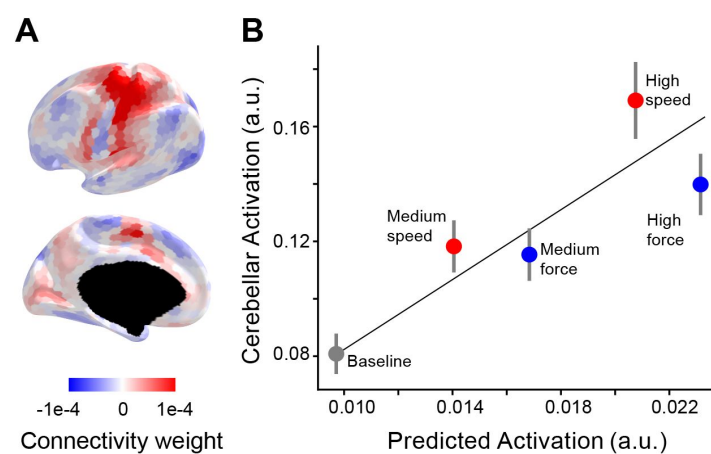


Figure 3.

Selective recruitment of cerebellum for fast alternating finger movements.

A) Average connectivity weights from a group-level connectivity model (Ridge regression, MDTB, task set A) for the cerebellar right-hand area shown on inflated surface of the left hemisphere. **B)** Average observed cerebellar activation (y-axis) plotted against average prediction from the connectivity model (x-axis). Resting baseline (located at 0,0) is not shown explicitly but included in the regression. The error bars indicate the standard error of the mean of the signed residuals.

1.94, $p = 0.035$). In summary, the increases in cerebellar activity for speed outstripped the activity increases for force, even when we accounted for differences in activity for the two conditions in neocortical input regions.

Alternative connectivity models

We recognize that our results depend on the connectivity model used to predict cerebellar activity. To ensure that our findings were robust, we replicated the results using 2 additional connectivity models. First, we used an L1-regularized model, which resulted in sparser connectivity weights. In our previous study, we found that this model performed only slightly worse in predicting left-out data compared to the L2-regularized model (King et al., 2023 [↗](#)). Second, we used a connectivity model that was trained on the entire MDTB dataset plus four additional large task-based datasets (Fusion model, see Methods). For both connectivity models, the predicted difference in the residual for the high-speed vs. high-force condition remained significant (L1 regression: $t_{15} = 2.373$, $p = 0.0315$, Fusion model: $t_{15} = 2.140$, $p = 0.0492$). Thus, consistent across various connectivity models, the results indicate selective recruitment of the cerebellum when the demands on finger coordination are increased relative to when the demands on force output are increased.

Working memory task

We conducted our initial test of the selective recruitment hypothesis using a motor task, for which we had a strong *a priori* prediction concerning the factors that lead to an upregulation of cortical input to the cerebellum. Having validated our approach here, we next turned to the cognitive domain, asking if our approach could help shed light on the functional contribution of the cerebellum to verbal working memory. Cerebellar activation, particularly in Lobules VI, Crus I and VIII is consistently observed in fMRI studies of working memory (Cohen et al., 1997 [↗](#); Courtney et al., 1997 [↗](#); D'Esposito and Postle, 2015 [↗](#); Nee et al., 2012 [↗](#)). Here we test if these cerebellar areas are especially recruited for a specific component process of working memory.

We implemented a digit span task in which participants memorized and subsequently recalled a sequence of visually presented digits (**Figure 4A** [↗](#)). Each trial began with a cue that signaled the recall direction (forward or backward) and the number of digits that had to be remembered (2, 4, 6). During the encoding phase, six digits were sequentially displayed from left to right at a rate of 1 digit/s. At the end of each 1 s presentation interval, the next digit was presented and the most recent digit either remained on the screen or was replaced by the hashtag symbol (#) if it had to be remembered. During the retrieval phase, all six digits had to be typed in on the keyboard, either backwards or forwards. Thus, while the memory load varied between 2 and 6 items, all conditions involved the presentation and production of 6 digits. On 25% of the trials (no-go) the trial was terminated at the start of retrieval phase. The inclusion of these encoding-only trials gave us separate estimates of activity related to the encoding and retrieval phases (see methods). In summary, we measured activity for 12 conditions (2 recall directions x 3 memory loads x 2 phases).

Figure 4B [↗](#) shows the error rate (trials with at least 1 wrong press) during the scanning session. As expected, error rates increased with memory load and were also higher in the backwards condition.

Consistent with previous imaging studies, the verbal working memory task led to high activity in the fronto-parietal network (Cohen et al., 1997 [↗](#); Courtney et al., 1997 [↗](#); D'Esposito and Postle, 2015 [↗](#); Nee et al., 2012 [↗](#); Owen et al., 2005 [↗](#)) during the encoding and retrieval phases (**Figure 5** [↗](#)). During the latter phase, we also observed activation in cortical motor areas, reflecting the response requirements of the task. Within the cerebellum, encoding and retrieval activated a superior region (lateral parts of lobule VI, extending to Crus I), as well as an inferior region (VIIb and VIIIa) (Chen and Desmond, 2005 [↗](#); Desmond et al., 1997 [↗](#)). As observed previously for

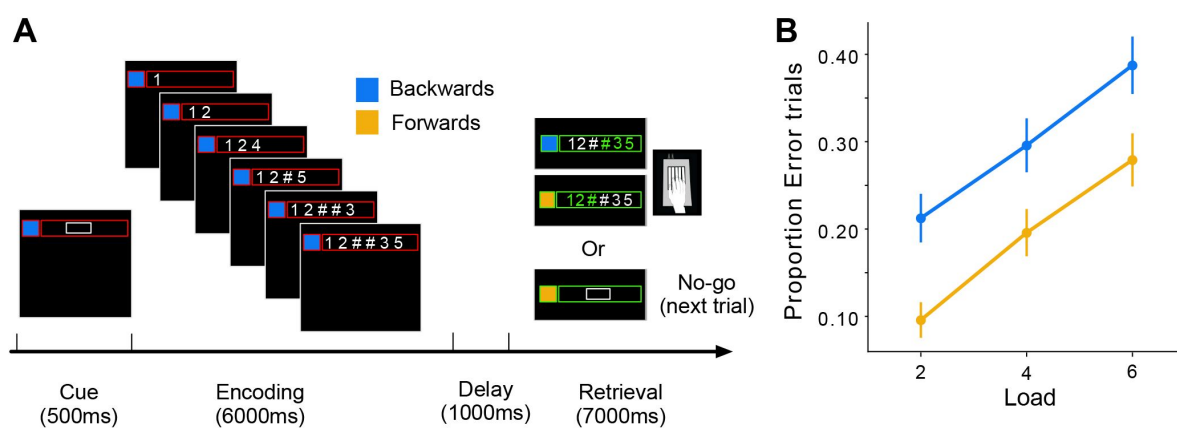


Figure 4.

The digit span task and behavioral performance.

a) Timeline of trial events. The cue signaled the recall direction (blue for backward and yellow for forward) and memory load (size of the white box indicated the number of memory digits) of the upcoming trial. During encoding, a new digit appeared every second and was replaced by the # symbol if it was a memory digit. After a 1 s delay, the task progressed to either the retrieval phase (Go trial) or skipped directly to the next trial (No-go trials). **b)** Proportion of error trials. Error bars indicate standard error of the mean across participants.

verbal working memory, the activity was more pronounced in the right than in the left cerebellar hemisphere (Desmond and Fiez, 1998 [↗](#)). In our symmetric functional atlas (Nettekoven et al., 2023 [↗](#)), the best corresponding functional region was right D3 (see *Methods, Region of interest*).

Cerebellar activity for encoding at high load is larger than predicted by task-invariant connectivity

To estimate which neocortical regions provide input to our cerebellar ROI (D3R), we again used the task-invariant model of cortico-cerebellar connectivity model (King et al., 2023 [↗](#)). The connectivity weights from the model (**Figure 6A** [↗](#)) suggest converging input from area 55b (located at the inferior end of the middle frontal gyrus), PEF (premotor eye field), area 6r (anterior to the primary motor cortex), and SCEF (dorsomedial frontal cortex (Glasser et al., 2016 [↗](#))). The model was used to predict activity in the cerebellar ROI for each condition and participant. After fitting a linear regression to account for scale differences between predicted and observed activations, we found that the predicted values matched the observed values relatively well at the individual level ($R^2 = 0.42$, $SE = 0.01$).

Turning to our test of selective recruitment, there was one clear deviation where the observed cerebellar activation was greater than predicted (**Figure 6B** [↗](#)): During encoding in the highest load condition. A repeated 1-factor measures ANOVA on the residuals across all 12 conditions found a systematic deviation across participants ($F_{11, 165} = 2.22$, $p = 0.0156$). When we analyze the residuals using a 2 (phase) $\times$ 2 (recall direction) $\times$ 3 (load) ANOVA, we found a significant 2-way interaction between load and phase ($F_{2, 30} = 4.38$, $p = 0.02$), but no significant effect of recall direction ($F_{1, 15} = 0.95$, $p = 0.34$). Thus, the results indicate selective recruitment of the cerebellum when the number of items to be encoded into working memory is high, an effect that holds for both the forward and backwards conditions.

As with the motor task, we repeated the analyses using two other cortico-cerebellar connectivity models. The same pattern of results was found with a significant difference across conditions (L1-regularized model trained on the MDTB dataset: $F_{11, 165} = 2.34$, $p = 0.0105$; L2-regularized model trained on 5 datasets: $F_{11, 165} = 2.55$, $p = 5.3 \times 10^{-3}$) due to a significant deviation from the predicted level of activation during encoding with a load of 6.

Discussion

Functional neuroimaging studies have shown that the human cerebellum is activated across a broad range of task domains. However, before inferring that the cerebellum contributes causally to these task, we need to consider that the BOLD signal in the cerebellar cortex is likely dominated by mossy-fiber input (Alahmadi et al., 2015 [↗](#), 2016 [↗](#); Gagliano et al., 2022 [↗](#); Mapelli et al., 2017 [↗](#); Mathiesen et al., 2000 [↗](#); Thomsen et al., 2004 [↗](#), 2009 [↗](#)), which in humans mostly carry information from neocortex. Thus, BOLD activation changes observed in the cerebellar cortex could be a consequence of the transmission of information through fixed anatomical connections. Given this possibility, it is problematic to assume that the activation implies a significant functional contribution, let alone make inferences about the nature of that contribution.

Our first experiment in the motor domain clearly illustrates this problem. We found highly significant increases in the cerebellar BOLD signal for increases in both rate and force. Using the inferential logic traditionally employed in neuroimaging, one might conclude that the cerebellum plays a functional role in regulating both parameters. However, clinical studies have shown that cerebellar pathology results in a marked impairment in the ability to produce fast alternating

Figure 5.

Average activation in the cortico-cerebellar network for working memory.

Group-averaged activation during the encoding (A) and retrieval (B) phases on an inflated representation of the left cerebral hemisphere (as in Figure 2). C, D) Group average activity during the two phases in the cerebellum. The D3R sub-region of the multi-demand network in the right cerebellar hemisphere was used in the main analysis (outlined in light gray).

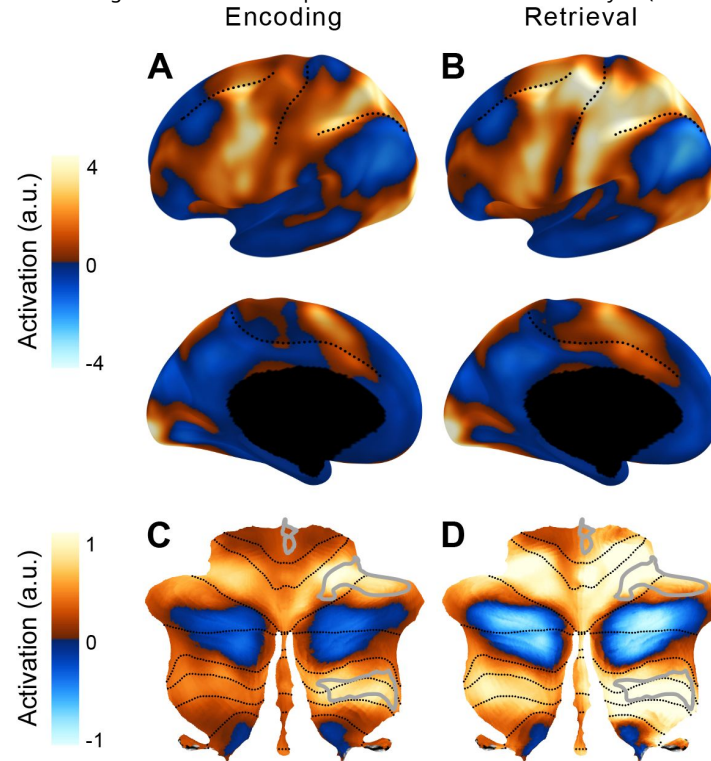
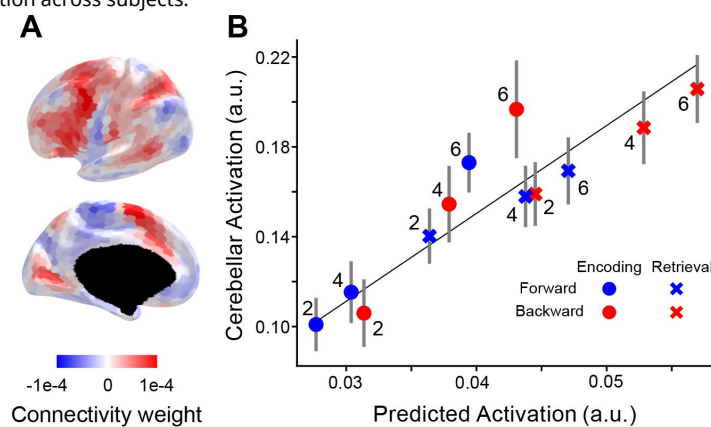


Figure 6.

Selective recruitment of cerebellum in digit span task.

A) Average connectivity weights from a group-level connectivity model for the cerebellar D3R region of interest. B) Average observed cerebellar activation (y-axis) plotted against average prediction from the connectivity model (x-axis). Line shows the best linear relationship between predicted and observed activity with an intercept of zero. Error bars show SEM of the signed residuals for each condition across subjects.



movements, but has little impact on maximal force generation (Mai et al., 1988 [↗](#)). This suggests that some observed change in cerebellar activity may not be functionally involved in controlling behavior.

To address these concerns, we first needed a strong null model: We used a cortico-cerebellar connectivity model (King et al., 2023 [↗](#)), which was optimized to predict cerebellar activity based only on neocortical activity patterns across a wide array of tasks; thus, it provides a prediction of the expected cerebellar activity for any task if all cerebellar activity was indeed caused by a fixed, task-invariant transmission of activity from neocortex. This prediction takes into account that some functional networks, such as the fronto-parietal and salience networks, occupy a relatively larger area of the cerebellum than of the neocortex (Buckner et al., 2011 [↗](#); Marek et al., 2018 [↗](#)), and that there will be variation in convergence across the cerebellar cortex. As shown in our previous study (King et al., 2023 [↗](#)), this model provides a good prediction of cerebellar activity across a broad range of tasks, including those not used in developing the model. This confirms that a large proportion of the observed variation of cerebellar activity across tasks can be accounted for by fixed functional connections between neocortical and cerebellar regions.

The central idea explored in the current paper is that systematic deviations from this null model would occur, if neocortical input was upregulated when the cerebellum is required for a task (and/or downregulated when it is not). We first tested this selective recruitment hypothesis in the motor domain, where we had the strong a-priori prediction: If input to the cerebellum is gated in a task-specific manner, then it should be upregulated during the production of fast alternating finger movements as compared to the production of high forces with the same fingers. This was indeed the case; activity in the cerebellar hand area increased more for increasing speed than force, even when activity in the neocortical hand areas was approximately matched across conditions. These results provide clear evidence for task-dependent gating (Cole et al., 2021 [↗](#)).

This phenomenon now offers a new, more stringent criterion to infer functional involvement of the cerebellum: Rather than focusing on activation for a given task per se, we can now test if a cerebellar area is selectively recruited for a task. If the input to the cerebellum is upregulated in a task-specific manner, then the observed cerebellar activity should be larger than predicted using the simultaneously observed neocortical activity and a task-invariant connectivity model. We can apply this approach to cognitive and social task to provide new insights into the contribution of the cerebellum in these domains.

As a first application, we chose to investigate cerebellar activation during a working memory task. Previous studies have consistently shown deficits in verbal working memory in cerebellar patients (Ilg et al., 2013 [↗](#); Kansal et al., 2017 [↗](#); Ravizza et al., 2006 [↗](#)). The exact nature of these deficits, however, is still a matter of considerable debate. Here we tested whether a variation in memory load, recall direction (requiring reversal of digits during backward recall), task phase (encoding vs. retrieval), or some combination of these factors, would lead to selective recruitment of the cerebellum. We found strong activity in cerebellar working memory regions across all conditions, with the retrieval phase for six items recalled in reversed order leading to most activation. Following the traditional neuroimaging inference approach, these results would be taken as evidence for cerebellar involvement in item manipulation during retrieval.

Our new analysis, however, highlights that the corresponding neocortical working memory region also showed the highest activation level during this condition and, importantly, that the cerebellar activity in this condition was well predicted using a task-invariant connectivity model. In contrast, the connectivity-based analysis identified the encoding of 6 items into working memory (for both forward and backward retrieval) as a condition for which the observed cerebellar activity outstripped the prediction by the null-model. This suggests that the cerebellum has a special role in encoding larger item sets into working memory. Further experiments using other working-memory tasks and a more detailed manipulation of encoding, maintenance and manipulation

processes will be required to precisely pinpoint the functional contribution of the cerebellum in this domain. Nonetheless, the current findings already provide important constraints on the cerebellar role in working memory, demonstrating the utility of our selective-recruitment approach for studying cerebellar function in cognition.

In terms of using this new approach to investigate cerebellar function, there are a number of important methodological factors to consider. First, the analysis heavily depends on the connectivity model that is used to predict the cerebellar activity. We addressed this issue by considering multiple models, including variations that allowed for more or less convergence (King et al., 2023 [↗](#)). We also showed that the results hold when using a model that is trained on a larger number of datasets (Nettekoven et al., 2023 [↗](#)), a step that improved the overall predictive accuracy of the approach (see **Supplementary Figure 1** [↗](#)). To be perfectly clear, we do believe that the task-invariant connectivity model is, as all models, ultimately *wrong* (Box, 1976 [↗](#)). Indeed, our current study shows clear and systematic deviations from the model's prediction. Nonetheless, we consider it to be a *useful* model, in that it can serve as a strong null hypothesis, one that can be used to test for task-specific upregulation of activity. Therefore, our approach will benefit from further improvements of the model, such that it approximates the average cortico-cerebellar connectivity as closely and representatively as possible.

Second, the connectivity model, as it is currently constructed, does not predict the absolute level of cerebellar activity, but rather activity for one condition relative to other conditions. This limitation arises from the fact that the absolute magnitude of the BOLD signal in the cerebellum depends on many measurement-related factors, and the fact that we need to apply relatively heavy regularization to obtain good model performance. We therefore need to estimate the linear relationship between predicted and observed activity for each participant separately. Thus, our approach currently relies on the comparison to control conditions that activate similar neocortical regions to comparable extent, but recruit the cerebellum to a lesser degree.

Thirdly, cortico-cerebellar connectivity is of course bi-directional. In our model, we do not model the influence of the cerebellum on neocortical activity, mediated through projections from the deep cerebellar nuclei to the thalamus. The simple reason for this decision is that cerebellar activity does not reflect the output firing of the Purkinje cells (Caesar et al., 2003 [↗](#); Thomsen et al., 2004 [↗](#), 2009 [↗](#)) and that, in contrast to cerebellar activity, cortical activity is determined by many other sources, including powerful cortico-cortical connectivity (King et al., 2023 [↗](#)).

Finally, our approach does not allow us to conclude that the cerebellum is *not* necessary for a when selective recruitment is *not* observed. Our approach simply shows that much of cerebellar activity can be fully accounted for by a task-invariant transmission of information from the neocortex, raising the possibility that this observed activity is an epiphenomenon of cortical input. Indeed, it would be very surprising if task-dependent gating was so complete that we would not see any activity in the cerebellar circuits that receive input from activated cortical regions. Given this, we should expect some cerebellar activity even when the cerebellum makes minimal contributions to task performance (as observed in the force condition). Overall, we believe that showing task-specific violations on a task-invariant connectivity model provides much stronger evidence for a specific cerebellar role in a task than the mere presence of activity.

An important question for future study centers on elucidating the neurophysiological mechanisms that underlie task-dependent gating of cortical input to the cerebellum. One obvious candidate for gating are the pontine nuclei which integrate descending signals from different neocortical areas with feedback signals from the cerebellum (Schwarz and Thier, 1999 [↗](#)). The cellular properties of pontine neurons are ideal for gating input signals in a state-dependent manner (Schwarz et al., 1997 [↗](#)). Alternatively, gating could be achieved via modulation within the granule cell layer itself, perhaps via recurrent loops involving inhibitory Golgi cells (Maex and Schutter, 1998 [↗](#)). Violations of our connectivity model may also be caused by increased climbing fiber input under

specific task conditions. Finally, gating may already occur in the neocortex: A recent study (Park et al., 2022 [↗](#)) showed more recruitment of neocortical neurons that project to the pons when controlling the spatial aspects of joystick manipulation, and more recruitment of neurons that project intra-cortically or to the striatum when controlling movement amplitude. Because the neocortical BOLD signal reflects the activity of both neuronal populations, pontine-projecting neurons may be more engaged during fast alternating movements, even though the fMRI activity is the same as during the production of high forces.

Whichever combination of mechanisms is responsible for our observed effect, task-dependent gating of inputs to the cerebellum would be highly adaptive from a metabolic standpoint (Attwell and Iadecola, 2002 [↗](#)): the costly mossy-fiber system would be most activated when cerebellar computation is required. For us as researchers, this gating phenomena offers a promising new keyhole that may allow us to unlock the use of fMRI for testing cerebellar contributions across cognitive tasks (Diedrichsen et al., 2019 [↗](#)).

Methods

Participants

All participants gave informed consent under an experimental protocol approved by the institutional review board at Western University. None of the participants reported a history of neurological or psychiatric disorders or current use of psychoactive medications. A total of 21 participants started the experiment. Of these, 4 participants were not scanned because of poor performance during the behavioral training session. The remaining 17 participants performed the tasks inside the scanner. The data for one participant was excluded due to an incidental finding. Therefore, the analyses were based on the data from 16 participants (8 female, 8 male, mean age = 25, std age = 2)

Apparatus and Stimuli

Participants used a custom-made 5-key finger keyboard to perform the finger tapping and digit span tasks. A force transducer, located under each key (FSG15N1A, Honeywell Sensing and Control; dynamic range, 0–25 N), continuously recorded the isometric force exerted by each finger at a rate of 500 Hz. We recalibrated each sensor (no force applied) at the beginning of each run to correct for drift. The applied force was continuously displayed to the participants in form of 5 short horizontal bars that moved along the vertical axis proportional to force exerted by each finger (Figure 1A [↗](#): applied forces)

Procedure

Finger tapping task

Each trial was randomly selected from one of five conditions (Figure 1B [↗](#)). In all conditions, the response interval lasted for 6 s and participants were instructed to adopt a rate to distribute their responses evenly across this interval. For the Baseline condition, the target force was 2.5 N, and the instructed number of presses was 6 (i.e., optimal performance is 1 response/s). For the medium and high force conditions, the target force was either 6 N or 10 N, with the target number of presses fixed at 6. For the medium and high-speed conditions, the target number of presses was 10 or 18, with the target force fixed at 2.5 N.

A trial started with a short cueing phase (500 ms) during which two numeric characters (3 and 4) were presented on the screen, instructing the participant to tap with the right middle and ring finger. The required force level was indicated by a gray box that extended from 80% to 120% of

the trial's target force (**Figure 1A**, target force area), and the required number of presses by either 6, 10, or 18 small gray squares (**Figure 1A**, instructed speed).

After the 500-ms cueing phase, the two rectangles framing the digits turned from white to green, signaling to the participants to perform alternating finger presses. A horizontal green line (**Figure 1A**, timer) started growing from left to right, indicating the passing of time. A press was registered when the force exceeded 80% of the target force (lower bound of the target force area). At this point, the force area changed color from gray to green and the color of the corresponding press square changed. When the force level returned to <1N, the force area color changed back to gray.

After the response phase, participants received performance feedback. If the participant made the required number (± 2) of alternating movements and completed the set of responses within 4 to 6 seconds, they received visual feedback indicating they had earned 4 points. This response time window was relatively liberal, because our main focus was not to match speeds exactly, but to get sufficient variation across conditions. All other outcomes were considered errors and were not rewarded (0 points). If the average exerted force for the trial exceeded 120% of the target force, the experimenter provided verbal feedback, asking the participant to press with less force. The message "TOO FAST" was displayed if total movement time was shorter than 4s or if the number of produced presses exceeded the instructed number by more than two. The message "TOO SLOW" was displayed if the number of produced presses by the end of the 6 s interval was 3 or more below the instructed number of presses. Visual feedback (points or error message) remained on the screen for 500 ms. After a delay of 500 ms (inter-trial-interval), the next trial began with the appearance of the next cue.

Digit span task

Each trial started with a short cuing phase (500 ms), during which a red frame was presented outlining where the digits would appear along with a colored square on the left side that specified the recall direction (orange=forward recall; blue=backward recall). A white box within the red frame outlined the digits that would have to be remembered (2,4, or 6). After the cue phase, a 6 s encoding phase started. Six digits were presented sequentially (1 s/digit) from left to right. The digits were drawn randomly (with replacement) from the set 1-5. The digits in the white box changed to a # symbol after 1 s; the other digits remained on the screen. For load 2 and load 4, the white box always encompassed the digits in the middle of the sequence (e.g., 13##45 or 1###5).

The encoding phase ended after the 1 s display time of the last digit and was followed by an additional 1 s delay. Following this, the procedure followed one of two paths. On No-Go trials, the screen blanked and 500 ms later, a new trial started with the cueing phase. These trials were included to be able to separate the activity associated with memory encoding and retrieval phases (see fMRI first-level analysis). On Go trials, the frame surrounding the digits turned green, indicating the start of the retrieval phase. Participants were instructed to press the key linked to the digit (1: thumb, 2: index, 3: middle, 4: ring, 5: pinky), either from memory or, for loads 2 and 4, from the visible digits on the screen. For forward trials (orange square), the participant was instructed to produce the responses to match the order observed in the encoding phase. For backwards trial (blue square), the participant was to reverse the sequential order of the digits, producing the right-most digit (last cued during encoding) first. For both conditions, the retrieval phase lasted for 7 seconds in total, giving participants enough time to complete the response (based on pilot work). To roughly match the speed of responding between the very easy (forward load 2) and very difficult (backwards load 6) conditions, participants were instructed to evenly space their responses across the 7 s retrieval period.

Participants received visual feedback immediately after each response. If the response was correct, the corresponding hashtag or digit turned green, if incorrect, red. Only one response was allowed for each item. At the end of the retrieval phase, participants received additional feedback for 500 ms summarizing trial performance (+4: all correct; +3: 1 error; +2: 2 errors; 0: otherwise). This point system was selected to encourage participants to attempt to recall each item.

Experimental sessions

Each participant completed two sessions, a practice session conducted outside the scanner and a test session conducted in the scanner. Each session involved five runs of the finger tapping task interleaved with five runs of the digit span task. Each run of the finger tapping task consisted of 5 repetitions of each of the 5 conditions with the order randomized (total of 25 trials/run, approx. 5 min/run). Each run of the digit span task consisted of 3 Go trials and 1 No-Go trial for each of the six conditions (3 Set sizes x 2 Recall Directions) with the order fully randomized (total of 24 trials/run, approx. 8 min/run). The practice session was completed between 3 and 10 days prior to the scanning session.

Image acquisition

MRI data was acquired on a 3T Siemens Prisma at the Center for Functional and Metabolic Mapping (CFMM) at Western University. A high-resolution whole brain anatomical MPRAGE image was acquired at the beginning of the scanning session (voxel size = 1 mm^3 , Field-of-view = $25.6 \times 25.6 \times 25.6\text{ cm}^3$). Whole-brain functional images were acquired using an echo-planar imaging sequence with TR = 1000 ms, TE = 30 ms, voxel size = $2.5 \times 2.5 \times 3\text{ mm}^3$, Field-of-view = $20.8 \times 20.8 \times 20.8\text{ cm}^3$, 48 slices, P to A phase encoding direction, with multi-band acceleration factor = 3 (interleaved) and in-plane acceleration factor = 2. Gradient echo field maps were acquired to correct for distortions due to B0 inhomogeneities (acquisition parameters: voxel size = $3 \times 3 \times 3\text{ mm}^3$, Field-of-view = $24 \times 24 \times 24\text{ cm}^3$). Physiological signals of heartbeat and respiration were recorded online during each functional run. Each functional run of the finger tapping task lasted ~5 minutes (260 volumes) and each run of the digit span task lasted for ~8 minutes (412 volumes).

fMRI data processing

We used tools from SPM12 (Friston et al., 1994 [DOI](#)) and custom written code in MATLAB 2018b to process the functional and anatomical data. We defined an individual coordinate system for each subject by setting the origin of the anatomical image to the approximate location of the anterior commissure. Anatomical images were segmented into gray matter, white matter, csf, and skull. Functional images were corrected for head motion using the six-parameter rigid body transformation and were then co-registered to the individual anatomical image. The mean functional image and the results of anatomical segmentation were used to generate a gray matter mask for functional images. Slice timing correction, smoothing, and group normalization were not applied at this stage.

fMRI first-level analysis

A first-level general linear model (GLM) was fit to the time series data of each run separately using SPM12. For the motor dataset, each condition was modeled as a separate regressor using a 6 s boxcar covering the response interval, convolved with a canonical hemodynamic response function (HRF). Error trials (approx. 5% of all trials) were modeled as one single regressor in the GLM and this regressor was discarded from further analysis.

For the working memory task, the encoding phase was modeled using a 7 s boxcar including 6 s of digit sequence display and the 1 s delay. The retrieval phase was modeled using a separate 7 s boxcar regressor covering the response interval. In go-trials the two regressors therefore followed each other immediately, leading to a substantial correlation after the convolution with the HRF. However, the inclusion of 25% no-go trials, for which only the encoding regressor was present, de-

correlated the encoding and retrieval regressors sufficiently to enable stable and accurate estimate of the two processes. For analysis of imaging data, we chose to include all trials, including trials in which the participants made an error. We justify this given that there was no evidence indicating that any participant ceased trying to do the task; as such, it is reasonable to assume that trials resulting in an error engaged the same memory processes.

Beta weights estimated by the first level GLM were divided by residual-root-mean-square image, resulting in normalized activity estimates for each voxel, condition, and run. Rest was not modeled explicitly but served as an implicit baseline. Functional and anatomical data were transformed into a cortical and cerebellar atlas using a unified code framework (available at https://github.com/DiedrichsenLab/Functional_Fusion).

Cerebellar normalization

The cerebellum was isolated from the rest of the brain and segmented into white and gray matter using the Spatially Unbiased Infratentorial Template (SUIT) toolbox (Diedrichsen, 2006), followed in some cases by hand correction. Cerebellar white matter and gray matter probabilistic maps were deformed simultaneously into SUIT atlas space using a non-linear deformation algorithm (Ashburner, 2007). The deformation was applied to both anatomical images, and the normalized beta weights from the first level GLM. Before normalization, the isolation mask was applied to discard the influence of adjacent inferior and occipital neocortical areas. For visualizations, the functional maps were projected onto a flat representation of the cerebellum (Diedrichsen and Zotow, 2015) using the SUIT toolbox.

Neocortical normalization

For each participant, the anatomical image was used to reconstruct neocortical white matter and pial surface using Freesurfer (Fischl, 2012). Reconstructed surfaces were inflated to a sphere and registered to the fsLR 32k node template (Van Essen et al., 2011) using a sulcal-depth map and local curvature. Neocortical activity patterns were projected onto these surfaces by averaging the activation values of voxels touching the line between corresponding vertices of the individual white matter and pial surface.

Region of Interest (ROI) selection

For both datasets, we used a new symmetric functional atlas of the human cerebellum (Nettekoven et al., 2023) that integrates data from 7 large task-based datasets. The regions within this parcellation were estimated using a hierarchical Bayesian approach, with the constraint that the boundaries between regions were symmetric in the left and right hemispheres. For the motor dataset, we focused on right M3, a subregion of the motor domain that shows high selectivity for right hand movements. For the working memory dataset, we focused on right D3, a subregion of the multi-demand network that showed the clearest response to verbal digit span and verbal N-back tasks in the training data for the atlas.

Connectivity model

We used task-invariant models of cortico-cerebellar connectivity to estimate the activity pattern in the cerebellar region of interest given the activity pattern in the cerebral cortex (King et al., 2023). This served as the null model from which we could evaluate the deviations in activity patterns, the test of the selective recruitment hypothesis. The models were trained on a large dataset with N=24 subjects, each of whom was scanned for ~6 hours using two sets of tasks spanning a large range of motor and cognitive domains (multi-domain task battery, MDTB; King et al., 2019). Each task set was performed in two sessions. For each participant, the neocortical surface was subdivided using regular icosahedron parcellations of different granularities, resulting in $P = 80-1848$ parcels. The normalized activity estimates (see first-level analysis) for all N conditions were then averaged within each parcel and collected into a $N \times P$ matrix. These

neocortical activations served as the predictors in the model ($\mathbf{X}$). The normalized activity estimates for the cerebellum were extracted in SUIT space at an isotropic resolution of 3 mm, resulting in an $N \times Q$ (29×6918) matrix ($\mathbf{Y}$). We estimated the $P \times Q$ matrix of connectivity weight ($\mathbf{W}$) by minimizing the square error of the linear regression model $\mathbf{Y} = \mathbf{X} \mathbf{W} + \mathbf{E}$. To regularize this underspecified estimation problem, we employed either L1 regularization (Lasso) or L2-regularization (Ridge regression). Hyperparameters were tuned using 5-fold cross-validation within the training data (see [King et al., 2023](#) [for details](#)).

The models were trained on the first task set (29 task conditions) and evaluated on the second task set of the MDTB (32 different task conditions acquired from the same participants). Predictive accuracy of the model was defined as the Pearson correlation between the observed and predicted response profile of each voxel across the tasks. For the present paper, we selected the model with the highest predictive accuracy across subjects, a ridge-regression model with a regularization parameter of $\lambda = \exp(8)$ and 1848 neocortical parcels/predictors. We also used the Lasso model ($\lambda = \exp(-5)$, 1848 neocortical parcels), to assess the generality of the results.

Finally, we also tested an improved connectivity model that was obtained by integrating data from five task-based datasets (including MDTB), totaling 376 task conditions, 87 subjects, and 383 hours of imaging data ([Nettekoven et al., 2023](#) [for details](#)). These new connectivity models were optimized and estimated on the individual subject level within each dataset, using L2-regularized regression (Ridge), and then averaged across all subjects and datasets. The data from the current study were not included in the derivation of these connectivity models.

To generate the predicted activity pattern, we used group-averaged connectivity weights for each voxel in the cerebellar region of interest. We extracted the functional neocortical data by averaging the individual data within each of the 1848 neocortical parcel. As rest was not modeled explicitly in our first-level analyses, we added the resting baseline as a row of zeros to both the cortical and cerebellar data. In this way, the connectivity model was required to simultaneously predict the differences between conditions and the differences between each condition and rest. The matrix of individual neocortical activations was multiplied using the group-averaged connectivity weights to arrive at an individual prediction for each cerebellar voxel.

Given that connectivity weights were derived from a different dataset with different subjects and different signal to noise ratios (SNR), we fitted a simple linear regression line for each participant between the observed cerebellar activation and the model predictions for the selected ROI. The slope of the line accounts for differences in SNR between the two datasets. Even though rest is not shown [Figure 3b](#) [and 6b](#) [for details](#), it was included as a datapoint at (0,0) for the regression analysis. The residuals from this regression analysis were used for statistical testing across participants.

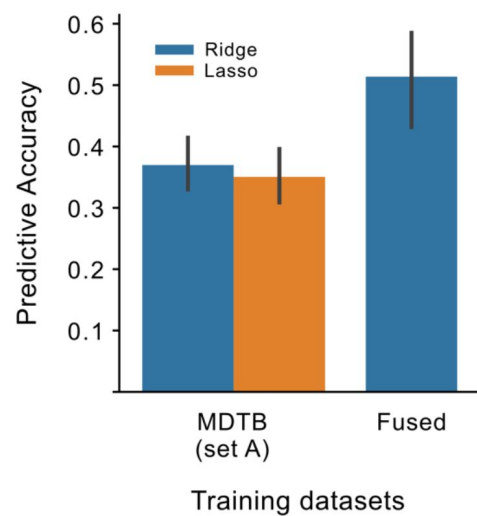
Supplementary figures

Acknowledgements

Funding

Canadian Institutes of Health Research (PJT 159520 to J.D.)

The Canada First Research Excellence Fund (BrainsCAN to Western University)



Supp. Figure. 1.

Connectivity models evaluation.

The connectivity model used in the main analysis was trained on task set A of the MDTB dataset using Ridge regression (King et al., 2023 [\[link\]](#)). As alternative connectivity models, we used Lasso regression on the same training set and a ridge regression model trained on 5 task-based dataset including MDTB (Zhi et al., 2023 [\[link\]](#)). Predictive accuracy was calculated as the cosine similarity between observed and predicted activity patterns of all the task conditions used in this paper.

Raynor Cerebellar Initiatives (to JD).

National Institute of Health (USA, NS116883 and NS105839 to R.I.)

Author contributions

Conceptualization: L.S., J.D., R.I.

Experiment Design: L.S., J.D.

Data collection: L.S.

Data analysis: L.S., J.D.

Writing—original draft: L.S., J.D.

Writing—review & editing: L.S., J.D., R.I., M.K., C.N.

Competing interests

R.I. is a co-founder with equity in Magnetic Tides, Inc. The other authors declare no competing interests.

Data and materials availability

The raw behavioral and imaging data for the cerebellum are also available on the data sharing repository <https://openneuro.org/> .

The analysis code is available at https://github.com/DiedrichsenLab/selective_recruitment .

References

Akgören N, Fabricius M, Lauritzen M. (1994) **Importance of nitric oxide for local increases of blood flow in rat cerebellar cortex during electrical stimulation** *Proc Natl Acad Sci U S A* **91**:5903–5907 <https://doi.org/10.1073/pnas.91.13.5903>

Alahmadi AAS, Pardini M, Samson RS, D’Angelo E, Friston KJ, Toosy AT, Gandini Wheeler-Kingshott CAM. (2015) **Differential involvement of cortical and cerebellar areas using dominant and nondominant hands: An FMRI study** *Hum Brain Mapp* **36**:5079–5100 <https://doi.org/10.1002/hbm.22997>

Alahmadi AAS, Samson RS, Gasston D, Pardini M, Friston KJ, D’Angelo E, Toosy AT, Wheeler-Kingshott CAM. (2016) **Complex motor task associated with non-linear BOLD responses in cerebro-cortical areas and cerebellum** *Brain Struct Funct* **221**:2443–2458 <https://doi.org/10.1007/s00429-015-1048-1>

- Allen G, Buxton RB, Wong EC, Courchesne E. (1997) **Attentional activation of the cerebellum independent of motor involvement** *Science* **275**:1940–1943 <https://doi.org/10.1126/science.275.5308.1940>
- Ashburner J. (2007) **A fast diffeomorphic image registration algorithm** *Neuroimage* **38**:95–113 <https://doi.org/10.1016/j.neuroimage.2007.07.007>
- Attwell D, Iadecola C. (2002) **The neural basis of functional brain imaging signals** *Trends Neurosci* **25**:621–625 [https://doi.org/10.1016/s0166-2236\(02\)02264-6](https://doi.org/10.1016/s0166-2236(02)02264-6)
- Box GEP (1976) **Science and Statistics** *J Am Stat Assoc* **71**:791–799 <https://doi.org/10.2307/2286841>
- Buckner RL, Krienen FM, Castellanos A, Diaz JC, Yeo BTT (2011) **The organization of the human cerebellum estimated by intrinsic functional connectivity** *J Neurophysiol* **106**:2322–2345 <https://doi.org/10.1152/jn.00339.2011>
- Caesar K., Gold L, Lauritzen M. (2003) **Context sensitivity of activity-dependent increases in cerebral blood flow** *Proc Natl Acad Sci U S A* **100**:4239–4244
- Kirsten Caesar, Thomsen K, Lauritzen M. (2003) **Dissociation of spikes, synaptic activity, and activity-dependent increments in rat cerebellar blood flow by tonic synaptic inhibition** *Proc Natl Acad Sci U S A* **100**:16000–16005 <https://doi.org/10.1073/pnas.2635195100>
- Chen SHA, Desmond JE (2005) **Cerebrocerebellar networks during articulatory rehearsal and verbal working memory tasks** *Neuroimage* **24**:332–338 <https://doi.org/10.1016/j.neuroimage.2004.08.032>
- Cohen JD, Perlstein WM, Braver TS, Nystrom LE, Noll DC, Jonides J, Smith EE (1997) **Temporal dynamics of brain activation during a working memory task** *Nature* **386**:604–608 <https://doi.org/10.1038/386604a0>
- Cole MW, Ito T, Cocuzza C, Sanchez-Romero R. (2021) **The Functional Relevance of Task-State Functional Connectivity** *J Neurosci* **41**:2684–2702 <https://doi.org/10.1523/JNEUROSCI.1713-20.2021>
- Cooper FE, Grube M, Von Kriegstein K, Kumar S, English P, Kelly TP, Chinnery PF, Griffiths TD (2012) **Distinct critical cerebellar subregions for components of verbal working memory** *Neuropsychologia* **50**:189–197 <https://doi.org/10.1016/j.neuropsychologia.2011.11.017>
- Courtney SM, Ungerleider LG, Keil K, Haxby JV (1997) **Transient and sustained activity in a distributed neural system for human working memory** *Nature* **386**:608–611 <https://doi.org/10.1038/386608a0>
- Desmond JE, Fiez JA (1998) **Neuroimaging studies of the cerebellum: language, learning and memory** *Trends Cogn Sci* **2**:355–362 [https://doi.org/10.1016/s1364-6613\(98\)01211-x](https://doi.org/10.1016/s1364-6613(98)01211-x)
- Desmond JE, Gabrieli JDE, Wagner AD, Ginier BL, Glover GH (1997) **Lobular Patterns of Cerebellar Activation in Verbal Working-Memory and Finger-Tapping Tasks as Revealed by Functional MRI** *J Neurosci* **17**:9675–9685 <https://doi.org/10.1523/JNEUROSCI.17-24-09675.1997>
- D’Esposito M, Postle BR (2015) **The cognitive neuroscience of working memory** *Annu Rev Psychol* **66**:115–142 <https://doi.org/10.1146/annurev-psych-010814-015031>

- Diedrichsen J. (2006) **A spatially unbiased atlas template of the human cerebellum** *Neuroimage* **33**:127–138 <https://doi.org/10.1016/j.neuroimage.2006.05.056>
- Diedrichsen J, King M, Hernandez-Castillo C, Sereno M, Ivry RB (2019) **Universal Transform or Multiple Functionality? Understanding the Contribution of the Human Cerebellum across Task Domains** *Neuron* **102**:918–928 <https://doi.org/10.1016/j.neuron.2019.04.021>
- Diedrichsen J, Wiestler T, Ejaz N. (2013) **A multivariate method to determine the dimensionality of neural representation from population activity** *Neuroimage* **76**:225–235 <https://doi.org/10.1016/j.neuroimage.2013.02.062>
- Diedrichsen J, Zotow E. (2015) **Surface-Based Display of Volume-Averaged Cerebellar Imaging Data** *PLoS One* **10** <https://doi.org/10.1371/journal.pone.0133402>
- Fischl B. (2012) **FreeSurfer** *Neuroimage* **62**:774–781 <https://doi.org/10.1016/j.neuroimage.2012.01.021>
- Friston KJ, Holmes AP, Worsley KJ, Poline J-P, Frith CD, Frackowiak RSJ (1994) **Statistical parametric maps in functional imaging: A general linear approach** *Hum Brain Mapp* **2**:189–210 <https://doi.org/10.1002/hbm.460020402>
- Gagliano G, Monteverdi A, Casali S, Laforenza U. (2022) **Non-Linear Frequency Dependence of Neurovascular Coupling in the Cerebellar Cortex Implies Vasodilation–Vasoconstriction Competition** *Cells* **11** <https://doi.org/10.3390/cells11061047>
- Glasser MF *et al.* (2016) **A multi-modal parcellation of human cerebral cortex** *Nature* **536**:171–178 <https://doi.org/10.1038/nature18933>
- Hallett M, Berardelli A, Matheson J, Rothwell J, Marsden CD (1991) **Physiological analysis of simple rapid movements in patients with cerebellar deficits** *J Neurol Neurosurg Psychiatry* **54**:124–133 <https://doi.org/10.1136/jnnp.54.2.124>
- Hokkanen LSK, Kauranen V, Roine RO, Salonen O, Kotila M. (2006) **Subtle cognitive deficits after cerebellar infarcts** *Eur J Neurol* **13**:161–170 <https://doi.org/10.1111/j.1468-1331.2006.01157.x>
- Howarth C, Gleeson P, Attwell D. (2012) **Updated energy budgets for neural computation in the neocortex and cerebellum** *J Cereb Blood Flow Metab* **32**:1222–1232 <https://doi.org/10.1038/jcbfm.2012.35>
- Howarth C, Peppiatt-Wildman CM, Attwell D. (2010) **The energy use associated with neural computation in the cerebellum** *J Cereb Blood Flow Metab* **30**:403–414 <https://doi.org/10.1038/jcbfm.2009.231>
- Ilg W, Christensen A, Mueller OM, Goerick SL, Giese MA, Timmann D. (2013) **Effects of cerebellar lesions on working memory interacting with motor tasks of different complexities** *J Neurophysiol* **110**:2337–2349 <https://doi.org/10.1152/jn.00062.2013>
- Jäncke L, Specht K, Mirzazade S, Peters M. (1999) **The effect of finger-movement speed of the dominant and the subdominant hand on cerebellar activation: A functional magnetic resonance imaging study** *Neuroimage* **9**:497–507 <https://doi.org/10.1006/nimg.1998.0426>

- Ji JL, Spronk M, Kulkarni K, Repovš G, Anticevic A, Cole MW (2019) **Mapping the human brain's cortical-subcortical functional network organization** *Neuroimage* **185**:35–57 <https://doi.org/10.1016/j.neuroimage.2018.10.006>
- Kansal K, Yang Z, Fishman AM, Sair HI, Ying SH, Jedynak BM, Prince JL, Onyike CU (2017) **Structural cerebellar correlates of cognitive and motor dysfunctions in cerebellar degeneration** *Brain* **140**:707–720 <https://doi.org/10.1093/brain/aww327>
- Kelly RM, Strick PL (2003) **Cerebellar loops with motor cortex and prefrontal cortex of a nonhuman primate** *J Neurosci* **23**:8432–8444
- King M, Hernandez-Castillo CR, Poldrack RA, Ivry RB, Diedrichsen J. (2019) **Functional boundaries in the human cerebellum revealed by a multi-domain task battery** *Nat Neurosci* **22**:1371–1378 <https://doi.org/10.1038/s41593-019-0436-x>
- King M, Shahshahani L, Ivry RB, Diedrichsen J. (2023) **A task-general connectivity model reveals variation in convergence of cortical inputs to functional regions of the cerebellum** *Elife* **12** <https://doi.org/10.7554/eLife.81511>
- Maex R, Schutter ED. (1998) **Synchronization of Golgi and Granule Cell Firing in a Detailed Network Model of the Cerebellar Granule Cell Layer** *J Neurophysiol* **80**:2521–2537 <https://doi.org/10.1152/jn.1998.80.5.2521>
- Mai N, Bolsinger P, Avarello M, Diener HC, Dichgans J. (1988) **Control of isometric finger force in patients with cerebellar disease** *Brain* **111**:973–998 <https://doi.org/10.1093/brain/111.5.973>
- Mapelli L, Gagliano G, Soda T, Laforenza U, Moccia F, D'Angelo EU (2017) **Granular Layer Neurons Control Cerebellar Neurovascular Coupling Through an NMDA Receptor/NO-Dependent System** *J Neurosci* **37**:1340–1351 <https://doi.org/10.1523/JNEUROSCI.2025-16.2016>
- Marek S *et al.* (2018) **Spatial and Temporal Organization of the Individual Human Cerebellum** *Neuron* **100**:977–993 <https://doi.org/10.1016/j.neuron.2018.10.010>
- Marvel CL, Desmond JE (2010) **Functional topography of the cerebellum in verbal working memory** *Neuropsychol Rev* **20**:271–279 <https://doi.org/10.1007/s11065-010-9137-7>
- Mathiesen C, Caesar K, Lauritzen M. (2000) **Temporal coupling between neuronal activity and blood flow in rat cerebellar cortex as indicated by field potential analysis** *J Physiol* **523**:235–246 <https://doi.org/10.1111/j.1469-7793.2000.t01-1-00235.x>
- McDougale SD, Tsay JS, Pitt B, King M, Saban W, Taylor JA, Ivry RB (2022) **Continuous manipulation of mental representations is compromised in cerebellar degeneration** *Brain* **145**:4246–4263 <https://doi.org/10.1093/brain/awac072>
- Nee DE, Brown JW, Askren MK, Berman MG, Demiralp E, Krawitz A, Jonides J. (2012) **A Meta-analysis of Executive Components of Working Memory** *Cereb Cortex* **23**:264–282 <https://doi.org/10.1093/cercor/bhs007>
- Nettekoven C, Zhi D, Shahshahani L, Pinho AL, Saadon-Grosman N, Diedrichsen J. (2023) **A hierarchical atlas of the human cerebellum for functional precision mapping** *bioRxiv* <https://doi.org/10.1101/2023.09.14.557689>

- O'Reilly JX, Beckmann CF, Tomassini V, Ramnani N, Johansen-Berg H. (2010) **Distinct and overlapping functional zones in the cerebellum defined by resting state functional connectivity** *Cereb Cortex* **20**:953–965 <https://doi.org/10.1093/cercor/bhp157>
- Owen AM, McMillan KM, Laird AR, Bullmore E. (2005) **N-back working memory paradigm: a meta-analysis of normative functional neuroimaging studies** *Hum Brain Mapp* **25**:46–59 <https://doi.org/10.1002/hbm.20131>
- Park J, Phillips JW, Guo J-Z, Martin KA, Hantman AW, Dudman JT (2022) **Motor cortical output for skilled forelimb movement is selectively distributed across projection neuron classes** *Science Advances* **8** <https://doi.org/10.1126/sciadv.abj5167>
- Peterburs J, Bellebaum C, Koch B, Schwarz M, Daum I. (2010) **Working memory and verbal fluency deficits following cerebellar lesions: relation to interindividual differences in patient variables** *Cerebellum* **9**:375–383 <https://doi.org/10.1007/s12311-010-0171-z>
- Petersen SE, Fox PT, Posner MI, Mintun M, Raichle ME (1989) **Positron emission tomographic studies of the processing of single words** *J Cogn Neurosci* **1**:153–170 <https://doi.org/10.1162/jocn.1989.1.2.153>
- Pleger B, Timmann D. (2018) **The role of the human cerebellum in linguistic prediction, word generation and verbal working memory: evidence from brain imaging, non-invasive cerebellar stimulation and lesion studies** *Neuropsychologia* **115**:204–210 <https://doi.org/10.1016/j.neuropsychologia.2018.03.012>
- Ravizza SM, McCormick CA, Schlerf JE, Justus T, Ivry RB, Fiez JA (2006) **Cerebellar damage produces selective deficits in verbal working memory** *Brain* **129**:306–320 <https://doi.org/10.1093/brain/awh685>
- Schwarz C, Möck M, Thier P. (1997) **Electrophysiological properties of rat pontine nuclei neurons In vitro. I. Membrane potentials and firing patterns** *J Neurophysiol* **78**:3323–3337 <https://doi.org/10.1152/jn.1997.78.6.3323>
- Schwarz C, Thier P. (1999) **Binding of signals relevant for action: towards a hypothesis of the functional role of the pontine nuclei** *Trends Neurosci* **22**:443–451 [https://doi.org/10.1016/S0166-2236\(99\)01446-0](https://doi.org/10.1016/S0166-2236(99)01446-0)
- Spraker MB, Corcos DM, Kurani AS, Prodoehl J, Swinnen SP, Vaillancourt DE (2012) **Specific cerebellar regions are related to force amplitude and rate of force development** *Neuroimage* **59**:1647–1656 <https://doi.org/10.1016/j.neuroimage.2011.09.019>
- Starowicz-Filip A, Prochwicz K, Kłosowska J, Chrobak AA, Myszka A, Bętkowska-Korpała B, Kwinta B. (2021) **Cerebellar Functional Lateralization From the Perspective of Clinical Neuropsychology** *Front Psychol* **12** <https://doi.org/10.3389/fpsyg.2021.775308>
- Thickbroom GW, Phillips BA, Morris I, Byrnes ML, Mastaglia FL (1998) **Isometric force-related activity in sensorimotor cortex measured with functional MRI** *Exp Brain Res* **121**:59–64 <https://doi.org/10.1007/s002210050437>
- Thomsen K, Offenhauser N, Lauritzen M. (2004) **Principal neuron spiking: neither necessary nor sufficient for cerebral blood flow in rat cerebellum** *J Physiol* **560**:181–189 <https://doi.org/10.1113/jphysiol.2004.068072>

- Thomsen K, Piilgaard H, Gjedde A, Bonvento G, Lauritzen M. (2009) **Principal cell spiking, postsynaptic excitation, and oxygen consumption in the rat cerebellar cortex** *Neurophysiol* **102**:1503–1512 <https://doi.org/10.1152/jn.00289.2009>
- van Es DM, van der Zwaag W, Knapen T. (2019) **Topographic Maps of Visual Space in the Human Cerebellum** *Curr Biol* **29**:1689–1694 <https://doi.org/10.1016/j.cub.2019.04.012>
- Van Essen DC, Glasser MF, Dierker DL, Harwell J, Coalson T. (2011) **Parcellations and Hemispheric Asymmetries of Human Cerebral Cortex Analyzed on Surface-Based Atlases** *Cereb Cortex* **22**:2241–2262 <https://doi.org/10.1093/cercor/bhr291>
- Van Overwalle F, D’aes T, Mariën P. (2015) **Social cognition and the cerebellum: A meta-analytic connectivity analysis** *Hum Brain Mapp* **36**:5137–5154 <https://doi.org/10.1002/hbm.23002>
- Zhi D, Shahshahani L, Nettekoven C, Pinho AL, Bzdok D, Diedrichsen J. (2023) **A hierarchical Bayesian brain parcellation framework for fusion of functional imaging datasets** *bioRxiv* <https://doi.org/10.1101/2023.05.24.542121>

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Reviewer #1 (Public Review):

This is an interesting and well-written paper reporting on a novel approach to studying cerebellar function based on the idea of selective recruitment using fMRI. The study is well-designed and executed. Analyses are sound and results are properly discussed. The paper makes a significant contribution to broadening our understanding of the role of cerebellum in human behavior.

In the revision, the authors did an excellent job in addressing my concerns.

<https://doi.org/10.7554/eLife.96386.2.sa1>

Reviewer #2 (Public Review):

Summary:

Shahshahani and colleagues used a combination of statistical modelling and whole-brain fMRI data in an attempt to separate the contributions of cortical and cerebellar regions in different cognitive contexts.

Strengths:

- * The manuscript uses a sophisticated integration of statistical methods, cognitive neuroscience and systems neurobiology.
- * The authors use multiple statistical approaches to ensure robustness in their conclusions.

* The consideration of the cerebellum as not a purely 'motor' structure is excellent and important.

Weaknesses:

* The assumption that cortical BOLD responses in cognitive tasks should be matched irrespective of cerebellar involvement does not cohere directly with the notion of 'forcing functions' introduced by Houk and Wise, suggesting the need for future work.

<https://doi.org/10.7554/eLife.96386.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

This is an interesting and well-written paper reporting on a novel approach to studying cerebellar function based on the idea of selective recruitment using fMRI. The study is well-designed and executed. Analyses are sound and results are properly discussed. The paper makes a significant contribution to broadening our understanding of the role of the cerebellum in human behavior.

We thank the reviewer for the positive assessment of our paper.

(1) While the authors provide a compelling case for the link between BOLD and the cerebellar cortical input layer, there remains considerable unexplained variance. Perhaps the authors could elaborate a bit more on the assumption that BOLD signals mainly reflect the input side of the cerebellum (see for example King et al., elife. 2023 Apr 21;12:e81511).

Our paper is based on the assumption that the cerebellar BOLD signal reflects solely the input to the cerebellum and does not reflect the changes in firing rates of Purkinje cells. This assumption relies on two lines of arguments: Studies that have directly looked at the mechanism of vasodilation in the cerebellum, and studies that try to infer the contributions of different neurophysiological mechanisms to overall cerebellar metabolism (Attwell and Iadecola, 2002).

Vasodilatory considerations: The mechanisms that causes vasodilation in the cerebellum, and hence BOLD signal increases, has been extensively studied: Electrical stimulation of mossy fibers (Gagliano et al., 2022; Mapelli et al., 2017), as well as parallel fibers (Akgören et al., 1994; Iadecola et al., 1996; Mathiesen et al., 1998; Yang and Iadecola, 1997) lead to robust increases in cerebellar blood flow. In contrast to the neocortex, the regulation of blood flow in the cerebellum depends nearly purely on the vasodilator Nitric Oxide (NO) (Akgören et al., 1994; Yang and Iadecola, 1997) with stellate cells playing a key role in the signaling cascade (Yang et al., 2000).

Electrical (Mathiesen et al., 2000) and pharmacological (Yang and Iadecola, 1998) stimulation of climbing fibers also leads to robust increases in blood flow. Simultaneous parallel and climbing fiber stimulation seems to combine sub-additively to determine the blood flow changes (K. Caesar et al., 2003).

Importantly, even dramatic changes in spiking rate of Purkinje cells do not lead to changes in vasodilation. For starters, parallel fiber stimulation leads to blood flow increases, even though the net effect on Purkinje cell firing is inhibitory (Mathiesen et al., 1998). More

importantly, complete inhibition of the Purkinje cell using a GABA agonist does not change baseline cerebellar blood flow (Kirsten Caesar et al., 2003). Conversely, even a 200-300% increase in simple (and complex) spike firing rate through application of a GABA antagonist does not show any measurable consequences for blood flow, even though it clearly increases the metabolic rate of oxygen consumption in the tissue (Thomsen et al., 2009, 2004).

In sum, this extensive set of studies clearly argues that the cerebellar blood flow response is mostly dictated by synaptic input, and that the firing rate of Purkinje cells does not influence vasodilation. Because the BOLD signal is caused by an supply of oxygen over and above the level of oxygen consumption, this would argue that increases in Purkinje cell firing would not lead to BOLD increases. What is less clear is the degree to which changes in BOLD signal during normal activity are determined by changes in mossy fiber or climbing fiber input. Disruption of either pathway leads to 60-70% reductions in the evoked blood flow response during whisker stimulation (Yang et al., 2000; Zhang et al., 2003) – but it remains unclear to what degree this reflects the distribution of contributions in the healthy animal, as these powerful disruptions may have a number of side-effects.

Metabolic considerations: To estimate the relative contributions climbing fiber / mossy fiber input to the variations in BOLD signal under natural conditions, it is useful to consider the contributions of different cerebellar processes to the overall metabolism of the cerebellum. Assuming an average firing rate of 40Hz for mossy fibers, ~3Hz for Granule cells, and 1Hz for climbing fibers, Howarth et al. (Howarth et al., 2012, 2010) estimated that the transmission from mossy fibers to granular cells, dominates the energy budget with 53%. The subsequent stage, encompassing the transfer of information from Granular cells to Purkinje cells, accounts for 32% of energy expenditure. In contrast, integration within Purkinje cells and the spiking (simple and complex) of these cells represents only 15% of the total energy consumption.

More important for the BOLD signal, however, are the activity-induced variations in metabolic consumption: Purkinje cells fire relatively constantly at a very high frequency (~50Hz) both during awake periods and during sleep (Shin et al., 2007). When providing a signal to the neocortex, firing rate decreases, actually lowering the metabolic demand. Climbing fibers normally fire at ~0.5 Hz and even during activity rarely fire much above 2Hz (Streng et al., 2017). In contrast, granule cells show a low firing rates during rest (typically <1hz) and can spike during activity well above 100Hz. Combined with the sheer number of granule cells, these considerations would suggest that the vast majority of the variation in metabolic demand are due to mossy fiber input and granule cell activity.

Overall, we therefore think it is likely that the main determinant of the cerebellar cortical BOLD signal is mossy fiber input and the transmission of information from mossy fibers to granule cells to Purkinje cells. We admit that the degree to which climbing fiber input contribute to BOLD signal changes is much less clear. We can be quite certain, however, that the firing rate of Purkinje cells does not contribute to the cerebellar BOLD signal, as even dramatic changes in the firing rate do not cause any changes in vasodilation. We have clarified our line of reasoning in the paper, and hope this more extensive response here will give the reader a better overview over the pertaining literature.

(2) The current approach does not appear to take the non-linear relationships between BOLD and neural activity into account.

Thank you for raising this concern. We did not stress this point in the paper, but one big advantage of our selective recruitment approach is that it is – to some degree- robust against non-linearities in the relationship between neural activity and BOLD signal. This is the case, as long as the shape of the non-linearity is similar in the cerebellum and the neocortex. The results of our motor task (Figure 3) provide a clear example of this: The BOLD signal both in

the neocortex and cerebellum incases non-linearly as a function of force – the increase from 2.5N to 6N (a 3.5N increase) is larger than the increase from 6N to 10N (a 4N increase). A similar non-linearity can be observed for tapping speed (6, 10 to 18 taps / s). However, within each condition, the relationship between cortical and cerebellar activity is nearly perfectly linear, reflecting the fact that the shape of the non-linearity for the cerebellum and cortex is very similar.

Most importantly, even if the non-linearity across the two structures is different, any non-linear relationship between neural activity and BOLD signal (of vasodilatory nature) should apply to different conditions (here force and speed increases) similarly. Therefore, if two conditions show overlapping activity levels (as observed for force and speed across medium and high levels, Figure 3), a offset between conditions cannot be caused by a non-linearity in the relationship of cortical and cerebellar activity. Because all conditions are subject to the same non-linearity, all points should lie on a single (likely monotonically increasing) non-linear function. Both for the motor and working memory task, the pattern of results clearly violates this assumption.

(3) The authors may want to address a bit more the issue of closed loops as well as the underlying neuroanatomy including the deep cerebellar nuclei and pontine nuclei in the context of their current cerebello-cortical correlational approach. But also the contribution of other brain areas such as the basal ganglia and hippocampus.

Cortical-cerebellar communication is of course bi-directional. As discussed in King et al., (2023), however, we are restricting our model to the connections from the neocortex to the cerebellum for the following reasons: First, cerebellar BOLD activity likely reflects mostly neocortical input (see our answer to pt. 1), whereas neocortical activity is determined by a much wider array of projections, including striato-thalamo-cortical and cortico-cortical connections. Secondly, the output of the cerebellum cannot be predicted from the BOLD signal of the cerebellar cortex, as it is unlikely that the firing rate of Purkinje cells contribute to cerebellar BOLD signal (see pt. 1). For these reasons we believe that the relationship between neocortical and cerebellar activity patterns is mostly dictated by the connectivity from cortex to cerebellum, and is therefore best modelled as thus. This is now more clearly discussed in a new paragraph (line 318-323) of the revised manuscript.

We are also ignoring other inputs to the cerebellum, including the spinal chord, the basal ganglia (Bhuvanandaram et al., 2022; Bostan and Strick, 2018) hippocampus (Froula et al., 2023; Watson et al., 2019), and amygdala (Farley et al., 2016; Jung et al., 2022; Terburg et al., 2024). In humans, however, the neocortex remains the primary source of input to pontine nuclei. Consequently, it stands as the main structure shaping activity within the cerebellar cortex. While it is an interesting question to what degree the consideration of subcortical structures can improve the prediction of cerebellar activity patterns, we believe that considering the neocortex provides a good first approximation.

Reviewer #1 (Recommendations):

(4) A few sentences to clarify the used models as was done in the King et al. (2024) paper may improve readability.

We have now added the sentences in the introduction (line 25ff):

To approach this problem, we have recently developed and tested a range of cortical-cerebellar connectivity models (King et al., 2023), designed to capture fixed, or task-invariant, transmission between neocortex and cerebellum. For each cerebellar voxel, we estimated a regularized multiple regression model to predict its activity level across a range of task conditions (King et al., 2019) from the activity pattern observed in the neocortex for the same

conditions. The models were then evaluated in their ability to predict cerebellar activity in novel tasks, again based only on the corresponding neocortical activity pattern. Two key results emerged from this work. First, while rs-FC studies (Buckner et al., 2011; Ji et al., 2019; Marek et al., 2018) have assumed a 1:1 mapping between neocortical and cerebellar networks, models which allowed for convergent input from multiple neocortical regions to a single cerebellar region performed better in predicting cerebellar activity patterns for novel tasks. Second, when given a cortical activation pattern, the best performing model could predict about 50% of the reliable variance in the cerebellar cortex across tasks (King et al., 2023).

(5) To what extent does this paper demonstrate the limitations of BOLD in neuroscientific research?

The primary objective of this study was to shed light on the problems of interpreting BOLD activation within the cerebellum. The problem that the BOLD signal mostly reflect input to a region is not unique to the cerebellum, but also applies (albeit likely to a lesser degree) to other brain structures. However, the solution we propose here critically hinges on three features of the cerebellar circuitry: a) the mossy fiber input for the cerebellar hemispheres mostly arise from the neocortex, b) the BOLD signal is likely dominated by this mossy fiber input (see pt. 1), and c) there is very little excitatory recurrent activity in the cerebellum, so output activity in the cerebellum does not cause direct activity in other parts of the cerebellum.

These features motivate us to use a directed cortex->cerebellum connectivity model, which does not allow for any direct connectivity within the cerebellum. While the same approach can also be applied to other brain structures, it is less clear that the approach would yield valid results here. For example, due the local excitatory recurrent connectivity within neocortical columns, the activity here will also relate to local processing.

(6) What if the authors reversed their line of reasoning as in that cerebellum activity is matched to map changes in cerebral cortical activity? Perhaps this could provide further evidence for the assumed directional specificity of the task-dependent gating of neocortical inputs.

Given (a) that the cerebellar BOLD signal tells us very little about cerebellar output signals (b) that there are many other input signals to the neocortex that are more powerful than cerebellar inputs, and c) that there strong cortical-cortical connections, we believe that this model would be hard to interpret (see also our answer to pt. 3).

Therefore, while the inversion of the linear task-invariant mapping between cortical and cerebellar activity is a potentially interesting exercise, it is unclear to us at this point what strong predictions we would be able to test with this approach.

(7) The statement that cerebellar fMRI activity may simply reflect the transmission of neocortical activity through fixed connections can be better explained. Also in the context of using the epiphenomenon (on page 11) in the paper. To what extent is the issue of epiphenomenon not a general problem of fMRI research?

We have rephrased the introduction of this idea (line 17):

This means that increases in the cerebellar BOLD signal could simply reflect the automatic transmission of neocortical activity through fixed anatomical connections. As such, whenever a task activates a neocortical region, the corresponding cerebellar region would also be activated, regardless of whether the cerebellum is directly involved in the task or not.

Epiphemonal activity: This is indeed a general problem in fMRI research (and indeed research that uses neurophysiological recordings, rather than manipulations of activity). Indeed, we have discussed similar issues in the context of motor activity in ipsilateral motor cortex (Diedrichsen et al., 2009). However, given that we only offer a possible approach to address this issue for the cerebellum (see pt. 5), we thought it best to keep the scope of the discussion focused on this structure.

Reviewer #2 (Public Review):

Summary:

Shahshahani and colleagues used a combination of statistical modelling and whole-brain fMRI data in an attempt to separate the contributions of cortical and cerebellar regions in different cognitive contexts.

Strengths:

The manuscript uses a sophisticated integration of statistical methods, cognitive neuroscience, and systems neurobiology.

The authors use multiple statistical approaches to ensure robustness in their conclusions.

The consideration of the cerebellum as not a purely 'motor' structure is excellent and important.

We thank the reviewer for their positive evaluation.

Weaknesses:

(1) Two of the foundation assumptions of the model - that cerebellar BOLD signals reflect granule cells > purkinje neurons and that corticocerebellar connections are relatively invariant - are still open topics of investigation. It might be helpful for the reader if these ideas could be presented in a more nuanced light.

Please see response to the comment 1 of Reviewer 1 for a more extensive and detailed justification of this assumption. We have now also clarified our rationale for this assumption better in the paper on line 10-14. Finally, we now also raise explicitly the possibility that some of the violations of the task-invariant model could be caused by selectively increase of climbing fiber activity in some tasks (line 340).

(2) The assumption that cortical BOLD responses in cognitive tasks should be matched irrespective of cerebellar involvement does not cohere with the idea of 'forcing functions' introduced by Houk and Wise.

We are assuming that you refer to the idea that cerebellar output is an important determinant of the dynamics (and likely also of the magnitude) of neocortical activity. We agree most certainly here. However, we also believe that in the context of our paper, it is justified to restrict the model to the connectivity between the neocortex and the cerebellum only (see reviewer 1, comment 3).

Furthermore, if increased cerebellar output indeed occurs during the conditions for which we identified unusually high cerebellar activity, it should increase neocortical activity, and bring the relationship of the cerebellar and cortical activity again closer to the predictions of the linear model. Therefore, the identification of functions for which cerebellar regions show selective recruitment is rather conservative.

Reviewer #2 (Recommendations):

(3) One of the assumptions stated in the abstract -- that the inputs to the cerebellum may simply be a somewhat passive relay of the outputs of the cerebral cortex -- has been challenged recently by work from Litwin-Kumar (Muscini et al., 2023 Nature Neuroscience), which argues for complex computational relationships between cortical pyramidal neurons, pontine nuclei and granule cells, which in turn would have a non-linear impact on the relationship between cortical and cerebellar BOLD. The modelling is based on empirical recordings from Wagner (2019, Cell) which show that the synaptic connections between the cortex and granule cells change as a function of learning, further raising concerns about the assumption that the signals inherent within these two systems should be identical. Whether these micro-scale features are indicative of the macroscopic patterns observed in BOLD is an interesting question for future research, but I worry that the assumption of direct similarity is perhaps not reflective of the current literature. The authors do speak to these cells in their discussion, but I believe that they could also help to refine the authors' hypotheses in the manuscript writ large.

We absolutely agree with your point. However, we want to make extremely clear here that our hypothesis (that the inputs to the cerebellum are a linear task-invariant function of the outputs of the cerebral cortex) is the Null-hypothesis that we are testing in our paper. In fact, our results show the first empirical evidence that task-dependent gating may indeed occur. In this sense, our paper is consistent with the theoretical suggestion of (Muscini et al., 2023).

You may ask whether a linear task-invariant model of cortical-cerebellar connectivity is not a strawman, given that is most likely incorrect. However, as we stress in the discussion (line 298-), a good Null-model is a useful model, even if it is (as all models) ultimately incorrect. Without it, we would not be able to determine which cerebellar activity outstrips the linear prediction. The fact that this Null-model itself can predict nearly 50% of the variance in cerebellar activity patterns across tasks at a group level, means that it is actually a very powerful model, and hence is a much more stringent criterion for evidence for functional involvement than just the presence of activity.

(4) Further to this point, I didn't follow the authors' logic that the majority of the BOLD response in the cerebellum is reflective of granule cells rather than Purkinje cells. I read through each of the papers that were cited in defense of the comment: "The cerebellar BOLD signal is dominated by mossy fiber input with very little contribution from the output of the cerebellar cortex, the activity of Purkinje cells" and found that none of these studies made this same direct conclusion. As such, I suggest that the authors soften this statement, or provide a different set of references that directly confirm this hypothesis.

Please see response to the comment 1, Reviewer 1. We hope the answer provides a more comprehensive overview over the literature, which DOES show that spiking behavior of Purkinje cells does not influence vasodilation (as opposed to mossy fiber input). We have now clarified our rationale and the exact cited literature on line 9-14 of the paper.

(5) Regarding the statement: "As such, whenever a task activates a neocortical region, we might observe activity in the corresponding cerebellar regions regardless of whether the cerebellum is directly involved in the task or not." -- what if this is a feature, rather than a bug? That is, the organisation of the nervous system has been shaped over phylogeny such that every action, via efference copies of motor outputs, is filtered through the complex architecture of the cerebellum in order to provide a feed-forward signal to the thalamus/cortex (and other connected structures). Houk and Wise made compelling arguments in their 1995 Cerebral Cortex paper arguing that these outputs (among other systems) could act as 'forcing functions' on the kinds of dynamics that arise in the

cerebral cortex. I am inclined to agree with their hypothesis, where the implication is that there are no tasks that don't (in some way) depend on cerebellar activity, albeit to a lesser or greater extent, depending on the contexts/requirements of the task. I realise that this is a somewhat philosophical point, but I do think it is important to be clear about the assumptions that form the basis of the reasoning in the paper.

This is an interesting point. Our way of thinking about cerebellar function does indeed correspond quite well to the idea of forcing functions- the idea that cerebellar output can “steer” cortical dynamics in a particular way. However, based on patient and lesion data, it is also clear that some cortical functions rely much more critically on cerebellar input than others. We hypothesize here that cerebellar activity is higher (as compared to the neocortical activity) when the functions require cerebellar computation.

We also agree with the notion that cerebellar contribution is likely not an all-or-none issue, but rather a matter of gradation (line 324ff).

(6) Regarding the logic of expecting the cortical patterns for speed vs. force to be matched -- surely if the cerebellum was involved more in speed than force production, the feedback from the cerebellum to the cortex (via thalamus) could also contribute to the observed differences? How could the authors control for this possibility?

Our model currently indeed does not attempt to quantify the contributions of cerebellar output to cortical activity. However, given that cerebellar output is not visible in the BOLD signal of the cerebellum (see reviewer 1, comment 1), we believe that this is a rational approach. As argued in our response to your comment 2, increased cerebellar output in the speed compared to the force condition should bring the activity relationship closer to the linear model prediction. The fact that we find increased cerebellar (as compared to neocortical) activity in the speed conditions, suggests that there is indeed task-dependent gating of cortical projections to the cerebellum.

Akgören N, Fabricius M, Lauritzen M. 1994. Importance of nitric oxide for local increases of blood flow in rat cerebellar cortex during electrical stimulation. *Proc Natl Acad Sci U S A* 91:5903–5907.

Attwell D, Iadecola C. 2002. The neural basis of functional brain imaging signals. *Trends Neurosci* 25:621–625.

Bhuvanasundaram R, Krzyspiak J, Khodakhah K. 2022. Subthalamic Nucleus Modulation of the Pontine Nuclei and Its Targeting of the Cerebellar Cortex. *J Neurosci* 42:5538–5551.

Bostan AC, Strick PL. 2018. The basal ganglia and the cerebellum: nodes in an integrated network. *Nat Rev Neurosci* 19:338–350.

Buckner RL, Krienen FM, Castellanos A, Diaz JC, Yeo BTT. 2011. The organization of the human cerebellum estimated by intrinsic functional connectivity. *J Neurophysiol* 106:2322–2345.

Caesar K., Gold L, Lauritzen M. 2003. Context sensitivity of activity-dependent increases in cerebral blood flow. *Proc Natl Acad Sci U S A* 100:4239–4244.

Caesar K., Thomsen K, Lauritzen M. 2003. Dissociation of spikes, synaptic activity, and activity-dependent increments in rat cerebellar blood flow by tonic synaptic inhibition. *Proc Natl Acad Sci U S A* 100:16000–16005.

Farley SJ, Radley JJ, Freeman JH. 2016. Amygdala Modulation of Cerebellar Learning. *J Neurosci* 36:2190–2201.

- Froula JM, Hastings SD, Krook-Magnuson E. 2023. The little brain and the seahorse: Cerebellar-hippocampal interactions. *Front Syst Neurosci* 17:1158492.
- Gagliano G, Monteverdi A, Casali S, Laforenza U, Gandini Wheeler-Kingshott CAM, D'Angelo E, Mapelli L. 2022. Non-linear frequency dependence of neurovascular coupling in the cerebellar cortex implies vasodilation-vasoconstriction competition. *Cells* 11:1047.
- Howarth C, Gleeson P, Attwell D. 2012. Updated energy budgets for neural computation in the neocortex and cerebellum. *J Cereb Blood Flow Metab* 32:1222–1232.
- Howarth C, Peppiatt-Wildman CM, Attwell D. 2010. The energy use associated with neural computation in the cerebellum. *J Cereb Blood Flow Metab* 30:403–414.
- Iadecola C, Li J, Xu S, Yang G. 1996. Neural mechanisms of blood flow regulation during synaptic activity in cerebellar cortex. *J Neurophysiol* 75:940–950.
- Ji JL, Spronk M, Kulkarni K, Repovš G, Anticevic A, Cole MW. 2019. Mapping the human brain's cortical-subcortical functional network organization. *Neuroimage* 185:35–57.
- Jung SJ, Vlasov K, D'Ambra AF, Parigi A, Baya M, Frez EP, Villalobos J, Fernandez-Frentzel M, Anguiano M, Ideguchi Y, Antzoulatos EG, Fioravante D. 2022. Novel Cerebello-Amygdala Connections Provide Missing Link Between Cerebellum and Limbic System. *Front Syst Neurosci* 16:879634.
- King M, Hernandez-Castillo CR, Poldrack RA, Ivry RB, Diedrichsen J. 2019. Functional boundaries in the human cerebellum revealed by a multi-domain task battery. *Nat Neurosci* 22:1371–1378.
- King M, Shahshahani L, Ivry RB, Diedrichsen J. 2023. A task-general connectivity model reveals variation in convergence of cortical inputs to functional regions of the cerebellum. *Elife* 12:e81511.
- Mapelli L, Gagliano G, Soda T, Laforenza U, Moccia F, D'Angelo EU. 2017. Granular layer neurons control cerebellar neurovascular coupling through an NMDA receptor/NO-dependent system. *J Neurosci* 37:1340–1351.
- Marek S, Siegel JS, Gordon EM, Raut RV, Gratton C, Newbold DJ, Ortega M, Laumann TO, Adeyemo B, Miller DB, Zheng A, Lopez KC, Berg JJ, Coalson RS, Nguyen AL, Dierker D, Van AN, Hoyt CR, McDermott KB, Norris SA, Shimony JS, Snyder AZ, Nelson SM, Barch DM, Schlaggar BL, Raichle ME, Petersen SE, Greene DJ, Dosenbach NUF. 2018. Spatial and Temporal Organization of the Individual Human Cerebellum. *Neuron* 100:977–993.e7.
- Mathiesen C, Caesar K, Akgören N, Lauritzen M. 1998. Modification of activity-dependent increases of cerebral blood flow by excitatory synaptic activity and spikes in rat cerebellar cortex. *J Physiol* 512 (Pt 2):555–566.
- Mathiesen C, Caesar K, Lauritzen M. 2000. Temporal coupling between neuronal activity and blood flow in rat cerebellar cortex as indicated by field potential analysis. *J Physiol* 523:235–246.
- Muscinelli SP, Wagner MJ, Litwin-Kumar A. 2023. Optimal routing to cerebellum-like structures. *Nat Neurosci* 26:1630–1641.
- Shin S-L, Hoebeek FE, Schonewille M, De Zeeuw CI, Aertsen A, De Schutter E. 2007. Regular patterns in cerebellar Purkinje cell simple spike trains. *PLoS One* 2:e485.
- Streng ML, Popa LS, Ebner TJ. 2017. Climbing Fibers Control Purkinje Cell Representations of Behavior. *J Neurosci* 37:1997.

Terburg D, van Honk J, Schutter DJLG. 2024. Doubling down on dual systems: A cerebellum–amygdala route towards action- and outcome-based social and affective behavior. *Cortex* 173:175–186.

Thomsen K, Offenhauser N, Lauritzen M. 2004. Principal neuron spiking: neither necessary nor sufficient for cerebral blood flow in rat cerebellum. *J Physiol* 560:181–189.

Thomsen K, Piilgaard H, Gjedde A, Bonvento G, Lauritzen M. 2009. Principal cell spiking, postsynaptic excitation, and oxygen consumption in the rat cerebellar cortex. *J Neurophysiol* 102:1503–1512.

Watson TC, Obiang P, Torres-Herraez A, Watilliaux A, Coulon P, Rochefort C, Rondi-Reig L. 2019. Anatomical and physiological foundations of cerebello-hippocampal interaction. *Elife* 8:e41896.

Yang G, Huard JM, Beitz AJ, Ross ME, Iadecola C. 2000. Stellate neurons mediate functional hyperemia in the cerebellar molecular layer. *J Neurosci* 20:6968–6973.

Yang G, Iadecola C. 1998. Activation of cerebellar climbing fibers increases cerebellar blood flow: role of glutamate receptors, nitric oxide, and cGMP. *Stroke* 29:499–507; discussion 507–8.

Yang G, Iadecola C. 1997. Obligatory role of NO in glutamate-dependent hyperemia evoked from cerebellar parallel fibers. *Am J Physiol* 272:R1155–61.

Zhang Y, Forster C, Milner TA, Iadecola C. 2003. Attenuation of activity-induced increases in cerebellar blood flow by lesion of the inferior olive. *Am J Physiol Heart Circ Physiol* 285:H1177–82.

<https://doi.org/10.7554/eLife.96386.2.sa3>